GATE DA (Data Science & AI) Core Maths Notes

A highly focused, exam-oriented study guide mapping Linear Algebra, Probability, Statistics, and Calculus constraints exactly to the official GATE syllabus requirements.

1. Linear Algebra (GATE DA Focus)

Linear algebra underpins dimensionality reduction (PCA), vector models, and transformations in Machine Learning. Expect high-weightage numerical problems from these topics.

Matrix Determinants & Inverse Short-Cuts

For a 2x2 matrix A = [[a, b], [c, d]], Det(A) = ad - bc. The inverse is given by A^{-1} = (1/Det(A)) * [[d, -b], [-c, a]]. Always check if Det(A) = 0 before computing inverse; if zero, the matrix is singular.

Eigenvalues and Eigenvectors (Core Concept)

Eigenvalues are scale factors by which space is stretched during transformation. Solve the characteristic equation: **Det(A - λI) = 0**.

Crucial Exam Properties:
1. Sum of Eigenvalues = Trace of Matrix (Sum of primary diagonal elements).
2. Product of Eigenvalues = Determinant of Matrix.

GATE Numerical Example (Eigenvalues):

If matrix A = [[3, 1], [1, 3]], find eigenvalues.
Characteristics equation: $(3-\lambda)(3-\lambda) - 1 = 0 \Rightarrow \lambda^2 - 6\lambda + 8 = 0$. Solving roots yields λ = 4 and λ = 2.

2. Probability & Statistics

Probability rules recommendation models, risk estimation, and bayesian classifiers. Focus directly on distributions and conditional workflows.

Conditional Probability & Bayes' Theorem

Bayes' theorem calculates posterior probability given prior knowledge of conditions. Crucial for classification problems.

Bayes' Theorem Formula:
*P(A|B) = [P(B|A) * P(A)] / P(B)*
Where P(B) is total probability: $P(B) = \Sigma P(B|A_i) * P(A_i)$

Random Variables & Probability Distributions

Discrete distributions map countable metrics (Binomial, Poisson); continuous distributions manage sliding scale values (Normal, Uniform).

Distribution	Key Formula (PDF/PMF)	Mean / Variance
Binomial	$P(X=k) = {}^{n!}/_{k!(n-k)!}\ p^k(1-p)^{n-k}$	Mean: np Var: $np(1-p)$
Poisson	$P(X=k) = (\lambda^k * e^{-\lambda}) / k!$	Mean: λ Var: λ
Normal (Z-score)	$Z = (X - \mu) / \sigma$	Mean: μ Var: σ^2

3. Calculus & Optimization

Calculus coordinates cost minimization routines (Gradient Descent) inside neural structures. Master derivatives and bounds verification.

Limits, Continuity & Single Variable Derivatives

Use L'Hopital's rule when evaluation limits yield indeterminate formats like 0/0 or ∞/∞. Take the derivative of numerator and denominator independently.

Maxima and Minima Criteria (Optimization)

Critical step for model training optimization workflows.

1. Find critical points by solving first derivative: ***f'(x) = 0****.*
2. Test second derivative f''(x) at these points:
- If $f''(x) > 0 \Rightarrow$ Local Minimum point.
- If $f''(x) < 0 \Rightarrow$ Local Maximum point.